



Quality of Tests · Mark Scheme

Further Statistics 1 Chapter 8: Quality of Tests. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Type I and Type II errors

- 1 (a) Rejecting H_0 when it is true B1; (b) not rejecting (accepting) H_0 when it is false B1.
- 2 $P(X \geq 10 \mid p = 0.3) = 1 - P(X \leq 9)$ M1; = 0.0480 A1.
- 3 Type II: $X \leq 9$ when $p = 0.5$ M1; $X \sim B(20, 0.5)$, $P(X \leq 9)$ M1; = 0.4119 A1.
- 4 (a) $P(X \leq 1) = 0.0404$, $P(X \leq 2) = 0.1247$ M1; critical region $X \leq 1$ A1; $P(\text{Type I}) = 0.0404$ B1. (b) $P(X \geq 2 \mid \lambda = 2) = 1 - 3e^{-2} = 0.5940$ B1.
- 5(a) $X \sim \text{Po}(6)$: $P(X \leq 1) = 0.0174$, $P(X \leq 2) = 0.0620$ M1; lower tail $X \leq 1$ A1; $P(X \geq 12) = 0.0201$, $P(X \geq 11) = 0.0426$ M1; upper tail $X \geq 12$ A1.
- 5(b) $0.0174 + 0.0201 = 0.0375$ (0.0374 from exact values) B1.
- 5(c) Type II: $2 \leq X \leq 11$ when $X \sim \text{Po}(8)$ M1; $P(X \leq 11) - P(X \leq 1)$ M1; = 0.8851 A1.
- 5(d) Power = $1 - 0.8851 = 0.1149$: low, so the test is unlikely to detect a change from 6 to 8 B1.

Section B – Power of a test

- 6 The probability of rejecting H_0 when it is false B1; power = $1 - P(\text{Type II error})$ B1.
- 7 Size = $P(X \leq 1 \mid p = 0.5) = \frac{11}{1024} = 0.0107$ B1; power = $P(X \leq 1 \mid p = 0.2) = 0.8^{10} + 10(0.2)(0.8)^9$ M1; = 0.3758 A1.
- 8 Power = $P(X \geq 3) = 1 - P(X \leq 2)$ M1; = $1 - e^{-\lambda} \left(1 + \lambda + \frac{1}{2}\lambda^2\right)$ A1; $\lambda = 3$: $1 - 8.5e^{-3} = 0.5768$ A1.
- 9 (a) $P(X \geq 10 \mid p = 0.4) = 0.0338$ B1. (b) Power = $1 - P(X \leq 9)$ for the given p M1; $p = 0.6$: 0.4032 A1; $p = 0.7$: 0.7216 A1.
- 10(a) A: $X \sim \text{Po}(2)$, $P(X \geq 5) = 0.0527$ B1; B: $Y \sim \text{Po}(4)$, $P(Y \geq 8) = 0.0511$ B1.
- 10(b) A: $X \sim \text{Po}(3)$ M1; power = $P(X \geq 5) = 0.1847$ A1; B: $Y \sim \text{Po}(6)$ M1; power = $P(Y \geq 8) = 0.2560$ A1.
- 10(c) The sizes are similar (about 5%), but test B has the greater power, so it is more likely to detect an increase B1; recommend B, although it takes twice as long B1.

Roar-level challenge

- 11 (a) $X_1 + X_2 \sim \text{Po}(6)$ under H_0 M1; size = $P(T \leq 2) = 25e^{-6} = 0.0620$ A1.
- (b) $T \sim \text{Po}(2\lambda)$, power = $e^{-2\lambda} \left(1 + 2\lambda + \frac{(2\lambda)^2}{2}\right)$ M1; = $e^{-2\lambda} (1 + 2\lambda + 2\lambda^2)$ A1; at $\lambda = 1$: $5e^{-2} = 0.6767$ B1.